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Skill 22.1 Exercise 1

For each of the following indicate whether the example is an aggregate statistic.

Example	Aggregate Statistic (yes/no)
Median household income in a city	yes
A single student's test score	no
A single household's income	no
Minimum stock price over the last 30 days	yes

What is an aggregate statistic?

An aggregate statistic is a way of creating a single number that describes a group of numbers.

Skill 22.2 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
4	77402	Donna	Blankenship	DB3807@gmail.com	stilettos	fabric	brown	289.0

Write code to answer the following,

What is the most expensive shoe	<code>shoe_fly.price.max()</code>
What is the least expensive shoe	<code>shoe_fly.price.min()</code>
What is the number of different colors of shoes	<code>shoe_fly.shoe_color.nunique()</code>
What is the average cost of shoes	<code>shoe_fly.price.mean()</code>
What are the different types of shoes	<code>shoe_fly.shoe_type.unique()</code>

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Skill 22.3 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
4	77402	Donna	Blankenship	DB3807@gmail.com	stilettos	fabric	brown	289.0

Write code to answer the following,

What is the most expensive shoe for each shoe type

```
shoe_fly.groupby('shoe_type').price.max()
```

What is the average cost of different colored shoes

```
shoe_fly.groupby('shoe_color').price.mean()
```

What is the median cost for shoes of different material

```
shoe_fly.groupby('shoe_material').price.mean()
```

What is the datatype returned for each statement above?

pandas Series

How can you convert each datatype into a dataframe?

```
shoe_fly.groupby('shoe_material').price.mean().reset_index() OR  
shoe_fly.groupby('shoe_material')[['price']].mean()
```

Skill 22.4 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
4	77402	Donna	Blankenship	DB3807@gmail.com	stilettos	fabric	brown	289.0

The following lambda function can be applied to return a value below which 25% of the data falls.

```
result = (lambda x: np.percentile(x, 25))
```

Use the lambda function above to answer the following,

What is the price threshold for the cheapest 25% of shoes of each type

```
result = (lambda x: np.percentile(x, 25))  
shoe_fly.groupby('shoe_type').price.apply(result)
```

What is the price threshold for the cheapest 10% of shoes of each material

```
result = (lambda x: np.percentile(x, 10))  
shoe_fly.groupby('shoe_material').price.apply(result)
```

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What is the price threshold for the cheapest 30% of shoes of each color	<code>result = (lambda x: np.percentile(x, 30)) shoe_fly.groupby('shoe_color').price.apply(result)</code>
What is the datatype returned for each statement above?	
<code>panda Series</code>	
How can you convert each datatype into a dataframe?	
<code>reset_index()</code>	

Skill 22.5 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
4	77402	Donna	Blankenship	DB3807@gmail.com	stilettos	fabric	brown	289.0

Write code to answer the following, store each result in a new dataframe.

What is the total number of shoes and each type and color?	<code>shoe_fly.groupby(['shoe_type', 'shoe_color']).id.count().reset_index()</code>
What is the average price of each shoe and color?	<code>shoe_fly.groupby(['shoe_type', 'shoe_color']).price.mean().reset_index()</code>
What is the price threshold for the cheapest 30% of shoes of each color and type?	<code>shoe_fly.groupby(["shoe_type", "shoe_color"]).price.apply(lambda x: np.percentile(x, 30)).reset_index()</code>

Skill 22.6 Exercise 1

Refer to the *Skill 22.5 Exercise 1*

For each statement, use pandas to create pivot table.

What is the total number of shoes and each type and color?	<code>pivot_table = pivot1_data.pivot(index='shoe_type', columns='shoe_color', values='id')</code>
What is the average price of each shoe and color?	<code>pivot_table2 = pivot2_data.pivot(index = 'shoe_type', columns = 'shoe_color', values = 'price')</code>
What is the price threshold for the cheapest 30% of shoes of each color and type?	<code>pivot_table3 = pivot3_data.pivot(index = 'shoe_type', columns = 'shoe_color', values = 'price')</code>